

Clinical Risk Assessment Report: Patient 69

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 28.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.28x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.3%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 2.52 [Top 4%] (+5.3%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.76 [Avg (74th %ile)] (+3.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.35 [Top 8%] (+2.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.06 [Top 15%] (+1.8%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.19 [Avg (26th %ile)] (+1.4%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

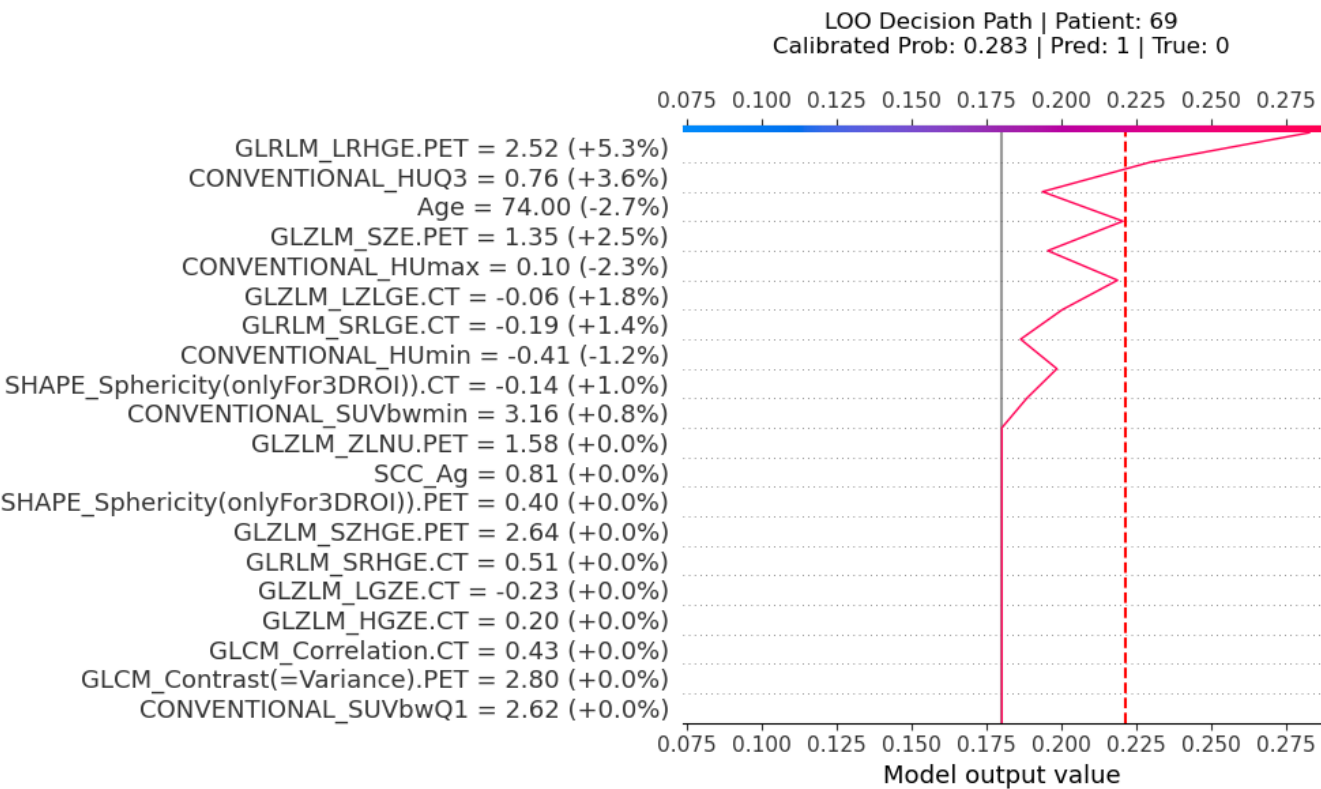
3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 74.0 [Top 13%] (-2.7%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.1 [Avg (49th %ile)] (-2.3%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.41 [Avg (32th %ile)] (-1.2%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

